Cheek Cell Culture + H2O2 Treatment Protocol v1.0.0

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This is a protocol drafted by combining methods used by two papers, one paper discusses the collection and culturing of human “cheek cells”/”buccal cells” (Michalczyk et al. 2004) and the other discusses treating epithelial cells with hydrogen peroxide to study immunology (Okahashi et al. 2014). This protocol was initially developed based on these papers, and was then modified after some testing. This protocol is meant to be repeatable and robust within the context of a science fair, while also remaining viable on a limited schedule.

Day 1. Cell Collection and Culture *~1 hour 30 minutes*

- *Required:*
    - *-70 ml* ***warm*** *Sterile Saline*
    - *1x 50ml Eppendorf tube*
    - *10ml serological pipettes (have 5ml and 20ml on-hand might be handy but not required)*
    - *20ml mouthwash*
    - *toothbrush*
    - *5-10ml* ***warm*** *DMEM + 10% FBS + 1% penecillin-streptomycin (“cell culture media”)*
    - *hemocytometer*
    - *trypan blue*
    - *micropipetter with pipette tips*
    - *microcentrifuge tubes or small glass test tubes (for mixing cells and trypan blue)*

1. Perform an oral rinse with 20 ml Listerine mouthwash for 30s, wait 5min before step 2
2. Vigorously brush teeth for 60s with a sterile toothbrush (using the same pressure used for normal brushing)
3. rinse mouth with 10ml sterile Saline for 60s and rinse the toothbrush with 10ml sterile Saline.
4. Pool together the two rinses
5. Wait 15 minutes, and repeat steps 1-4 for a total of two collections.
6. centrifuge rinses at 3000xg for 5 minutes, aspirate the supernatant and wash the pellet of cells by re-suspending in in 25ml sterile saline using a pipette.
7. Aspirate the supernatant again and re-suspend in 2ml cell culture media

8. look at the cells in a hemocytometer and calculate the total number of cells:
    1. Thuroughly vortex cell suspension and immediately pipette 50ul into a tube
    2. add 25ul clean cell culture media into the same tube
    3. add 25ul Trypan Blue into the same tube
    4. clean the hemocytometer and coverglass with lens paper
    5. place the coverglass over the dry hemocytometer
    6. vortex and immediately add 10ul of stained cells into the hemocytometer using the v-shaped notch on the side of the hemocytometer
    7. count the number of cells in one of the main squares
    8. multiply that number by 10,000 cells/ml
    9. multiply that number by 2 (because we diluted 50ul cells in 100ul final volume, our dilution factor is 2)
    10. checkout this website for more information: https://bitesizebio.com/13687/cell-counting-with-a-hemocytometer-easy-as-1-2-3/
9. Calculate the amount of media needed to achieve a concentration is $1x10^6$ cells per ml using the following formula:

    [cells/ml calculated from hemocytometer] * 2ml (from step 8) = Total # of cells collected

    [Total # of cells collected] / [$1x10^6$ cells/ml] = Final volume of media required

    [Final volume required] – 2ml (from step 8) = amount of media to add (<u>for next step</u>)

10. add media to achieve $1x10^6$ cells per ml
11. Culture the cells overnight in a shaker at 37° C @ 225rpm

Day 2. H_2O_2 + polyphenol treatment *~45 minutes*

-required:

- *5-10ml* ***warm*** *DMEM + 10% FBS + 1% penecillin-streptomycin ("cell culture media")*
- *~1ml sterile saline*
- *4x 15ml eppendorf tubes*
- *Hydrogen peroxide*
- *rosemary extract*
- *various pipettes*

1. Centrifuge cells at 3000xg for 5 minutes, resuspend in the same [Final volume] of cell culture media calculated the day before
2. allocate 4 equal volumes of cell culture into 15ml eppendorf tubes
3. There will be the following treatments: H_2O_2 only, Rosemary Extract (RE) only, H_2O_2 and RE, and one left untreated
4. Treat the H_2O_2 only and the H_2O_2+RE tubes with H_2O_2 (final concentration of 0.3% H_2O_2) for 10 minutes in 37° C shaker @ 225rpm
5. Centrifuge all 4 tubes at 3000xg for 5 minutes, resuspend in cell culture media
6. Dilute 100ul rosemary extract in 900ul sterile saline (1:10 dilute)
7. Treat the RE only and H_2O_2+RE tubes with a 1:100 ratio of diluted rosemary extract (i.e. if the cell culture volume is 500ul, treat with 5ul diluted RE)

    NOTE: 1:10 dilution followed by a 1:100 dilution makes a <u>final dilution of 1:1000</u>
8. Culture the cells overnight in a shaker at 37° C @ 225rpm

Day 3. Cell viability assay and data collection *~1.5 hours*

-required:

- *hemocytometer*
- *trypan blue*
- *micropipetter with pipette tips*
- *microcentrifuge tubes or small glass test tubes (for mixing cells and trypan blue)*

1. look at the cells in a hemocytometer and calculate the total number of living and dead cells:
    1. Thuroughly vortex cell suspension and immediately pipette 50ul into a tube
    2. add 25ul clean cell culture media into the same tube
    3. add 25ul Trypan Blue into the same tube
    4. clean the hemocytometer and coverglass with lens paper
    5. place the coverglass over the dry hemocytometer
    6. vortex and immediately add 10ul of stained cells into the hemocytometer using the v-shaped notch on the side of the hemocytometer
    7. count the number of blue cells (dead cells) and clear cells (living cells) in **2** of the main squares
    8. multiply that number by 10,000 cells/ml
    9. multiply that number by 2 (because we diluted 50ul cells in 100ul final volume, our dilution factor is 2)
    10. divide that number by 2 (because you counted 2 of the main squares)
    11. checkout this website for more information: https://bitesizebio.com/13687/cell-counting-with-a-hemocytometer-easy-as-1-2-3/

Sources

Michalczyk A, Varigos G, Smith L, Ackland ML. 2004. Fresh and cultured buccal cells as a source of mRNA and protein for molecular analysis. BioTechniques. 37(2):262–269. doi:10.2144/04372RR03.

Okahashi N, Sumitomo T, Nakata M, Sakurai A, Kuwata H, Kawabata S. 2014. Hydrogen Peroxide Contributes to the Epithelial Cell Death Induced by the Oral Mitis Group of Streptococci. Ratner AJ, editor. PLoS ONE. 9(1):e88136. doi:10.1371/journal.pone.0088136.